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DISTRIBUTION UTILITY MEET
DUM 2026

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DUM 2026 CONFERENCE SESSION

Scaling Digital Intelligence and Advanced Smart Meter Data Analytics

Theme Presentation

Presented By

GAUTAM KUMAR, CSO , INTELLISMART INFRA

Distribution Utility Meet | 27 - 28 October 2026 | www.dumindia.in



The Digital Foundation

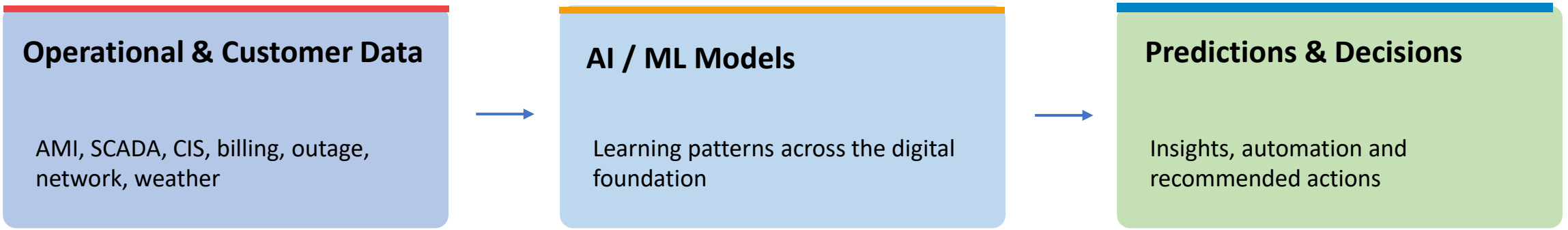
- Advanced Metering Infrastructure (AMI) & smart-grid deployments
- Enterprise-level IT / OT integration
- Renewable & distributed generation, energy storage
- Cloud and edge computing platforms

- Clean-energy transition, distributed generation, energy storage and the demand for better customer experience are now driving unprecedented digital disruption.
- Next-generation distribution utility is evolving into a digital, information-age, connected enterprise — spanning generation, the grid and the end customer.

The Intelligence Layer Over the Digital Foundation

Artificial Intelligence and Machine Learning will sit above this digital foundation

Processing large volumes of operational and customer data into predictions, insights and increasingly automated decisions



Classified as Internal

IMPORTANCE OF SMART METER DATA & AI (GRID RESILIENCE & FLEXIBILITY)

Unmanaged Data Risks

Blind spots hide massive aggregate technical and commercial losses across feeder networks.

Bidirectional Flows

Rooftop solar and EVs create complex micro-fluctuations below grid substations.

Predictive AI Role

24 billion daily interval reads require automated ML models for real-time stability.

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WHAT DISCOMS AND AMISPs ARE DOING WITH DATA

National AI Innovations

REC Powerthon showcased real-world applications (TNPDCL, MP East, Tata Power).

AMISP Pre-SAT Checks

AI enables OCR /Validation solution for Smart Meter MI , Communication Quality Assurance. More than 2 Lacs MI records validated

MSEDCL Vitaran Intelligence

Deployed AI platform for predictive maintenance and grid reliability enhancements.

UHBVN Revenue Recovery

Pilot (~50k sample) unlocked INR 26.16 crore (~$3M) in potential revenue from theft & unbilled consumers.

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RDSS STANDARD BIDDING DOCUMENT (SBD) CLAUSES

Clause 2.7.8 & 2.7.3(i)

Sole Custody & Ownership

Utility remains the sole custodian and absolute owner of all smart meter data.

Clauses 6.1 & 6.2.1

MDM Analytics Mandate

Meter Data Management systems must actively support advanced AI/ML-based reporting.

The Regulatory Gap

Commercial Framework

Silent on who funds AI infrastructure, owns trained models, and shares value between stakeholders.

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DPDP COMPLIANCE & SMART METER DATA SHARING

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Roles & Custodianship

DISCOM is Fiduciary, AMISP is Processor, consumer is Principal. DISCOM is sole custodian.

Forward-Looking Framework

Balances commercial data value with robust consumer safeguards for high-volume flows.

Privacy & Anonymization

Reports anonymized to remove traceability. Commercial sharing requires approval & opt-outs.

Statutory Exemptions

Consent exempt for DISCOM duties, law enforcement, and aggregated academic research.

High-Stakes Penalties

Strict non-compliance penalties: up to ₹250 Cr (security lapses), ₹200 Cr (unnotified breaches), ₹50 Cr (general violations).

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MOVING BEYOND PILOTS TO COMMERCIAL VIABILITY

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Physical vs. Digital Layer

Smart meters form sensor hardware; true value lives in software analytics and decision engines.

The Pilot Trap

High upfront and sustained infrastructural costs keep numerous AI meter data projects stuck in pilot stages.

The Solution

Move away from isolated proofs-of-concept and establish clear commercial terms for enterprise scale.

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WAY FORWARD FRAMEWORK

PILLAR 01

Commercial Framework

Cost sharing, IP ownership, value distribution, and DPDP/MeitY AI alignment.

PILLAR 02

SBD Change Requests

Utilize standard +20% Change Request mechanism within existing SBD contracts for rapid AI deployment.

PILLAR 03

Consolidating for Scale

Empower established AMISPs with operational infrastructure to drive enterprise analytics.

The real opportunity is not only to automate existing processes — it is to build predictive, flexible, customer-centric and self-optimizing framework for

High-Quality Data

Interoperable Architecture

Operational Discipline & Governance

For Reliability, Cybersecurity, Human Accountability and Deterministic Grid controls

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THANK YOU

For discussions/suggestions/queries email: ***dum@indiasmartgrid.org***

www.dumindia.in

Links/References (If any)

